

# MATH 127 Calculus for the Sciences

## Lecture 20

# Today's lecture

**Last time**

Continuity and continuity rules

**This time**

**Course note coverage** Section 3.1.3 and 3.1.4

Intermediate value theorem (IVT)

Classifying discontinuities

## Classifying discontinuity

Given a function  $f(x)$ , we can look at three things

- (1) the value at  $a$ :  $f(a)$
- (2) the left limit:  $\lim_{x \rightarrow a^-} f(x)$
- (3) the right limit:  $\lim_{x \rightarrow a^+} f(x)$

There are some possibilities when the function is not continuous

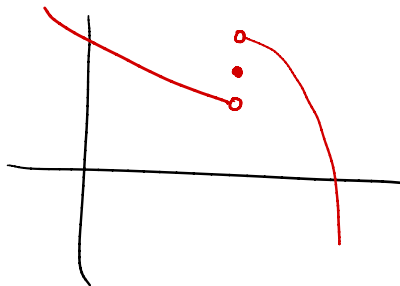
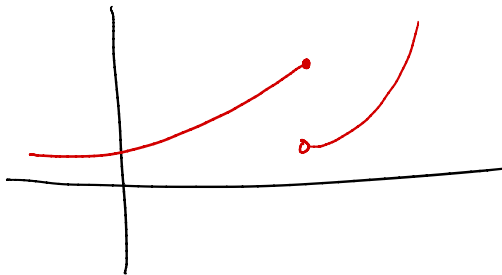
- (1)=(2), but (3) is a number different from them.
- (1)=(3), but (2) is a number different from them.
- (2)=(3), but (1) is a number different from them.
- (1)=(2), but (3) is  $\infty$ .
- (1)=(3), but (2) is  $\infty$ .
- (2)=(3), but  $a$  is not in the domain of  $f$ , so (1) does not exist.

## Jump discontinuity

- (1) the value at  $a$ :  $f(a)$   
(2) the left limit:  $\lim_{x \rightarrow a^-} f(x)$   
(3) the right limit:  $\lim_{x \rightarrow a^+} f(x)$

There are some possibilities when the function is not continuous

- (1)=(2), but (3) is a number different from them.
- (1)=(3), but (2) is a number different from them.

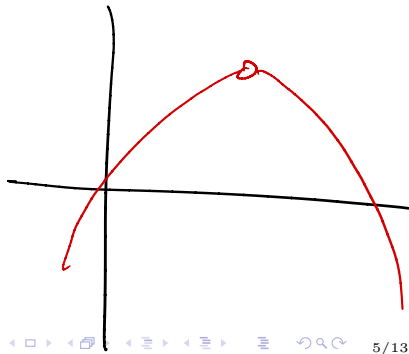
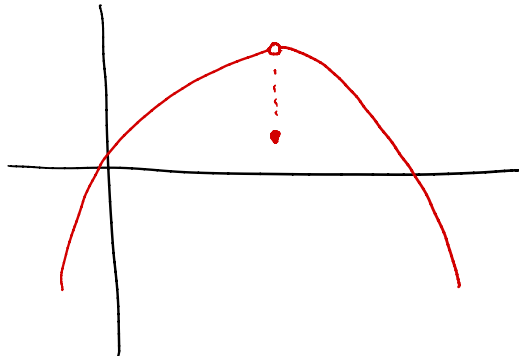


## Removable discontinuity

- (1) the value at  $a$ :  $f(a)$   
(2) the left limit:  $\lim_{x \rightarrow a^-} f(x)$   
(3) the right limit:  $\lim_{x \rightarrow a^+} f(x)$

There are some possibilities when the function is not continuous

- (2)=(3), but (1) is a number different from them.
- (2)=(3), but  $a$  is not in the domain of  $f$ , so (1) does not exist.

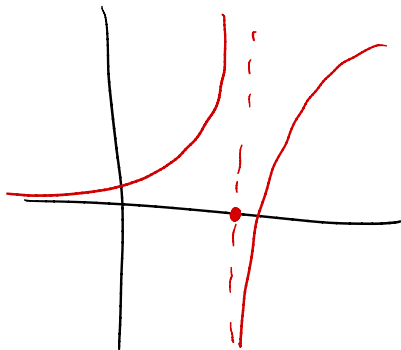
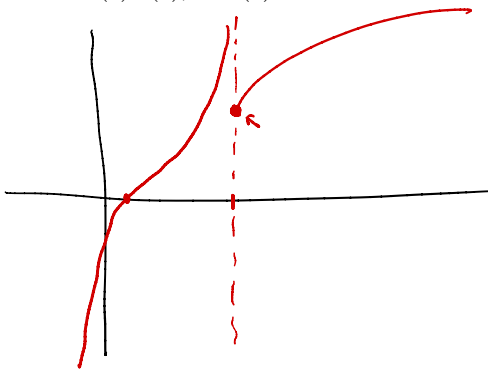


## Infinity discontinuity

- (1) the value at  $a$ :  $f(a)$
- (2) the left limit:  $\lim_{x \rightarrow a^-} f(x)$
- (3) the right limit:  $\lim_{x \rightarrow a^+} f(x)$

There are some possibilities when the function is not continuous

- (1)=(2), but (3) is  $\infty$ .
- (1)=(3), but (2) is  $\infty$ .



## Intermediate value theorem

Suppose there is a UFO:

- Moving smoothly in the air, up and down, every day.
- We know for sure that its height is 1000m at 7AM, and 2000m at 7PM.

Then I ask

- Do we know that the UFO has height 500m at some point? If so, when does it happen?  $\Delta$
- Do we know that the UFO has height 1000m at some point? If so, when does it happen?  $\circ, 7AM$
- Do we know that the UFO has height 1500m at some point? If so, when does it happen?  $\circ, ???$
- If I want to set a floating camera to catch the UFO, what heights can I set it to, to be guaranteed to catch the UFO?

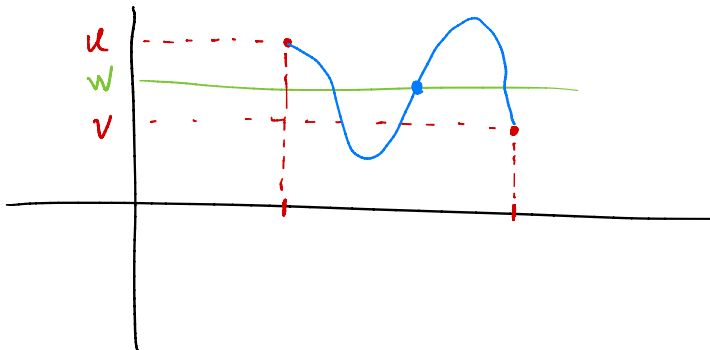
$$1000 \leq \text{height} \leq 2000$$

## Intermediate value theorem

### Conclusion:

If a thing has height  $U$  at some time, moves continuously and reaches height  $V$  at a later time, then:

for any  $W$  between  $U$  and  $V$ , the thing must reach height  $W$  within that period of time.



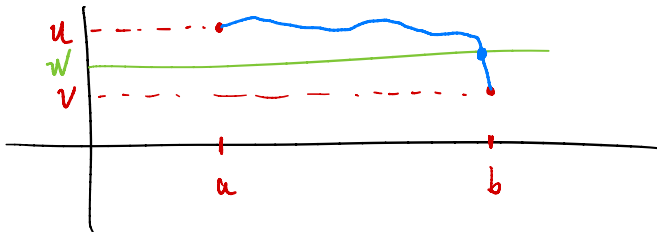
## Intermediate value theorem

### Conclusion:

If a thing has height  $U$  at some time, moves continuously, and reaches height  $V$  at a later time, then:  
for any  $W$  between  $U$  and  $V$ , the thing must reach height  $W$  within that period of time.

### Let us rephrase this:

If  $f(a) = U$  and  $f(b) = V$ , and  $f$  is continuous, then:  
for any  $W$  between  $U$  and  $V$ , then  $f$  must achieve an output of  $W$  within the interval  $[a, b]$



## Intermediate value theorem

### Conclusion:

If  $f(a) = U$  and  $f(b) = V$ , and  $f$  is continuous, then:  
for any  $W$  between  $U$  and  $V$ , then  $f$  must achieve an output of  $W$   
within the interval  $[a, b]$

### Rephrase again:

If  $f(a) = U$  and  $f(b) = V$ , and  $f$  is continuous, then:  
for any  $W$  such that

$$U < W < V$$

$W$  inside  $[U, V]$

there exists some  $c$  inside  $[a, b]$  such that



$$f(c) = W$$

## Intermediate value theorem

### Conclusion:

If  $f(a) = U$  and  $f(b) = V$ , and  $f$  is continuous, then:  
for any  $W$  such that

$$U < W < V,$$

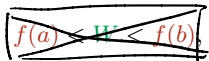
there exists some  $c$  inside  $[a, b]$  such that

$$f(c) = W.$$

### Final rephrasing:

#### Theorem (Intermediate value theorem (IVT))

If  $f$  is continuous on the interval  $[a, b]$ , then for any  $W$  such that


$$f(a) < W < f(b)$$

$W$  inside  $[f(a), f(b)]$

there exists  $c$  inside  $[a, b]$  such that

$$f(c) = W.$$

## Example

### Example

Does the polynomial  $x^3 + x + 1$  have a root in the interval  $[-1, 1]$ ?

**Example** This is the same asking: Does there exist a value  $c$  in the interval  $[-1, 1]$  such that the function  $f(x) = x^3 + x + 1$  satisfies  $f(c) = 0$ ?

1. by continuity rule,  $x^3 + x + 1$  is continuous on  $[-1, 1]$

2. Check:  $0$  in  $[f(-1), f(1)] = [(-1)^3 + (-1) + 1, 1^3 + 1 + 1]$   
 $= [-1, 3]$

- Yes, this is true.

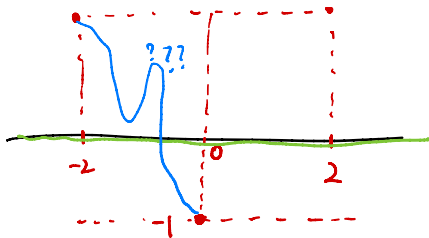
3. By IVT, there exists some  $c$  such that  $f(c) = 0$   
 So the polynomial  $x^3 + x + 1$  has a root at  $c$ .

## Example

### Example

Does the polynomial  $x^2 - 1$  have a root in the interval  $[-2, 2]$ ?

Use the intermediate value theorem to show this.



1.  $f(x) = x^2 - 1$  is continuous.

2. 0 is inside

$$[f(-2), f(0)] = [-1, 3]$$

↑      ↑

3. By IVT there exists  $c$   
inside  $[-2, 0]$   
that  $f(c) = 0$